

"Hello World"

```
System.out.println("Hello World")
```

String : ^{array} sequence of characters



[A-Z] [a-z] [0-9]

< @, \$, %, # . . . >

space, newline, new tab

"abc123"

"123"

123

"abc\$"

String or not?

✓

✓

✗ this is integer

✗

String
"123"

✓/s

Integer
123

On string, we apply operations like concatenation, length, slicing (substring) etc.

On integer, we apply arithmetic operations like +, -, /

String in Computers

Computer only understands binary (base 2)

↓
0 and 1

Human understandable num system

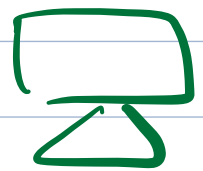
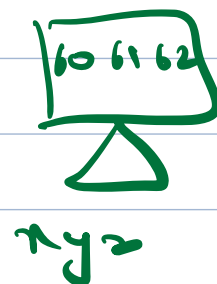
↓
digits 0-9 (base 10)

"xyz" → [a b c]

x → num a

y → num b

z → num c



b → 60
q → 61
r → 62

⋮
↓

We want mapping b/w characters and numbers to be uniform in every computer, that's how ASCII standard was born.

ASCII (American Standard Code for Information Interchange)

A standard data encoding format for electronic communication b/w computers.

ASCII assign standard numeric values to all characters (letters, numerals, special ...)

A → 65

B → 66

C → 67

⋮

Z → 90

a → 97

b → 98

c → 99

⋮

z → 122

0 → 48

1 → 49

2 → 50

⋮

9 → 57

10 → 58 ×

11 → 59 ×

A, Z
[65, 90]

$$90 - 65 = 26 \times$$

[6, 10]

$$10 - 6 = 4 \times$$

6, 7, 8, 9, 10 → 5

"10" → [49, 48]

"11" → [49, 49]

[a, b] Nos. in range from a to b
including a and b = $b - a + 1$

A²
[65, 90]

Nos. in range = $90 - 65 + 1$
= 26 ✓

a^z
[97, 122]

Nos. = $122 - 97 + 1$
= 26 ✓

ASCII Mapping → 128 characters
[0 - 127]

UTF-8, UTF-16 → Unicode Mapping

"\\ \\ \\ \\"

"Apuva said - '\\hi\\'"

//

"Coke // Tea"